

# Rohan Kallur

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## Skills

**Languages:** Python, TypeScript, JavaScript, SQL, Java, Solidity, C, C++, Rust

**AI/ML:** PyTorch, YOLO, OpenAI API, Claude API, Weights&Biases, Roboflow, Label Studio, NumPy, OpenCV, PIL, Pandas

**Backend:** FastAPI, PostgreSQL, Node.js, Django, REST APIs, MongoDB

**Frontend:** React, Next.js, Tailwind CSS, React Native, Zustand, Framer Motion

**Tools & Infra:** Git, Docker, AWS, Unix, CI/CD, Playwright, Mocha, Chai, Jupyter Notebook, Google Colab

## Work Experience

### Software Engineer

May 2026 – Present

AvaWatz

Remote

- Owned a 4-class incisor segmentation model end to end, hand-annotating ~4,000 masks across 1,007 images, auditing train/val leakage by perceptual hash, and fine-tuning YOLOv8l-seg in PyTorch to 0.93 mask mAP@50–95.
- Engineered an automated TMJ grid construction pipeline in Python (NumPy, OpenCV, PIL), benchmarking 5 unique axis computation methods across 140 panoramic x-rays from pre-segmented landmark coordinates, collaborating with the client's Chief Specialist Officer over three sessions until verified correct on every case.
- Built and shipped an end-to-end pipeline behind a clinical dashboard (React, TypeScript, FastAPI) — batch-processing uploads through segmentation and grid construction to dentist-facing results, deployed in beta.

### Software Development Intern

May 2024 – Aug 2024

Netki

Remote

- Developed an Ethereum smart contract (Solidity) for DeFi token verification, enforcing KYC compliance for digital assets; used Truffle and Ganache for local blockchain simulation.
- Refactored smart contract logic to reduce gas costs by 50%, optimizing event emissions, storage layout, algorithmic execution.
- Built automated test suites with Mocha/Chai, enabling reliable CI/CD integration for production deployments.
- Authored developer documentation for client integration of the contract, detailing function behavior and implementation steps used in production client workflows.

### Research Assistant

Jan 2023 – May 2023

Stevens Institute for Artificial Intelligence

Hoboken, NJ

- Built Python prototypes for Generative AI and NLP (Natural Language Processing) pipelines as part of an NSF-funded collaboration with Columbia and Syracuse — focused on extracting structured information from unstructured data.
- Engineered a OpenAI GPT-4 (beta) tool generating headlines from image datasets via vector embeddings, similarity metrics, and clustering algorithms.
- Fine-tuned models in PyTorch and built interactive tools for computational journalism applications.

## Projects

### NBA ReDrafter *redraft-room.onrender.com*

Jan 2026 – Present

- Designed and shipped a full-stack web app using React 19, TypeScript, FastAPI, and PostgreSQL — including drag-and-drop UI, shareable links, and screenshot export via Playwright headless browser automation.
- Engineered a Python data pipeline to scrape, merge, and ingest NBA draft data from multiple sources into PostgreSQL, matching records across sources by name and draft selection across 25 draft classes (2000–2025) and 2,500+ players.
- Architected and deployed a REST API (FastAPI) with endpoints for data retrieval, image proxying, and token-based ephemeral storage — designed for concurrency and optimized for low-latency responses in production on Render

### Portfolio Website *rohankallur.xyz*

Feb 2026 – Present

- Built and deployed a personal site (Next.js 16, React 19, TypeScript, Tailwind CSS v4) featuring a custom CS:GO-inspired case-unboxing entry animation (Framer Motion) revealing the site content.
- Built a contact form (Next.js API routes, Resend) with client-side validation and honeypot spam prevention, plus light/dark theming via custom React context.

## Education

### Stevens Institute of Technology

Sept 2021 – May 2025

Bachelor of Science in Computer Science, Minor in Finance

Hoboken, NJ

- GPA: 3.7/4.0 | *Magna Cum Laude* | Dean's List (All Semesters) | Upsilon Pi Epsilon (CS Honor Society)
- Relevant Coursework (CS):** Data Structures, Algorithms, Probability and Statistics, Linear Algebra, Database Management Systems, Operating Systems
- Relevant Coursework (Finance Minor):** Intro to Quantitative Finance, Intro to Derivatives, Portfolio Management, Corporate Finance, Financial Accounting